

Week 10 Tutorial

Estimating required rates of return for projects

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Presentation

- The presentations will all be held in **Week 12**.
- Each group will have a maximum of **8 minutes**.
- The presentation order will be determined by **drawing lots** for the groups.
- All group members must present; no one is exempt.
- The Maximum number of slides is **15**
- The structure and the contents of the slides and any other aspects of the slides (colour etc) are **your group's decision**.
- Email me your complete slides in PPT before the presentation class.
- If your group needs to present in Week 11 instead, please let me know in advance (by Sunday, May 11).

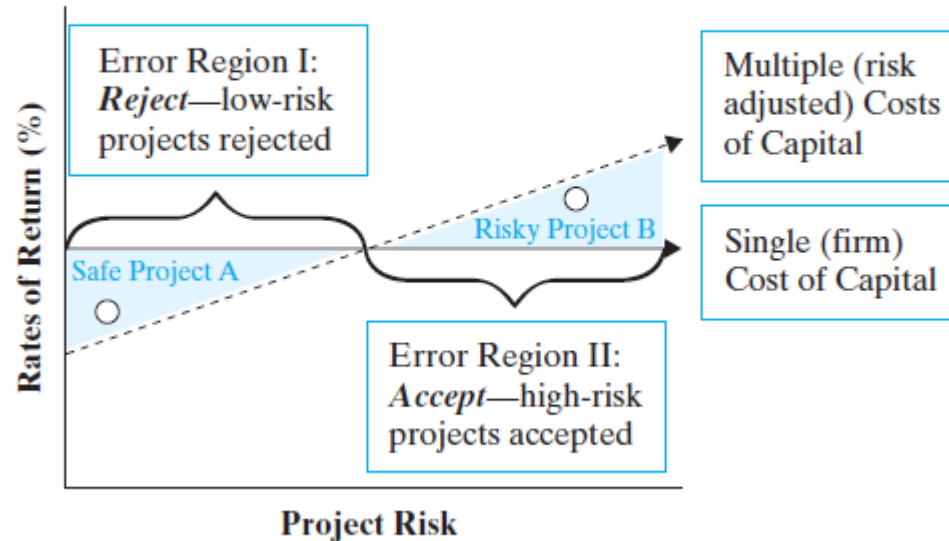
Agenda

- 5-4
- 5-7
- 5-8
- 5-9

Review 1: Multiple discount rates

- The limitation of a single discount rate (firm WACC)

Figure 5-1 Using a Single Cost of Capital Can Bias Investment Decisions Toward Risky Projects



Review 1: HPI

Convincing Your Skeptical Boss—Low-Risk Projects Can Be Good Investments

To illustrate the advantages of using multiple discount rates, consider the situation faced by a hypothetical high-tech manufacturing company, Huson Packer Inc. (HPI), which generates most of its revenues from technology-driven products that generate relatively high rates of return. This is a risky business and, as such, the firm is all-equity-financed, with a cost of equity and company cost of capital equal to 12%. Some of the firm's engineers have designed a new laser technology and have proposed that the firm invest \$300 million to launch a line of laser printers to commercialize the technology. Based on an analysis similar to what was described in Chapters 2 and 3, the engineers estimate that the investment will generate an expected internal rate of return of 10.5%. Although the expected return is lower than the firm's WACC of 12%, it should be noted that the project is also less risky than the firm's core business. In particular, the beta for the printer business is only .80, while the beta of HPI's core business is 1.40.

After reviewing the proposal, HPI's CEO responded, "I can't understand how it makes sense to undertake an investment that generates a 10.5% return when our shareholders expect a return on their investment of 12%." How would you convince your

Review 1: HPI

skeptical boss that the investment makes sense? As we explain below, to understand how an investment that returns 10.5% makes sense for a firm with a WACC of 12%, we must understand the distinction between the marginal cost of capital and the average cost of capital.

Let's assume that the expected return on the market portfolio is 10% and the risk-free rate is 5%, so that the market risk premium is $10\% - 5\% = 5\%$. Under these conditions, HPI estimates its cost of equity using the capital asset pricing model (CAPM) to be 12% (i.e., $5\% + 1.4 \times 5\%$). Now let's assume that the printer business will constitute 20% of the value of HPI if the investment is undertaken. With the printer business, HPI's new beta will become a weighted average of the beta for the printer business and its core business, which will constitute 80% of the value of the company. Consequently, HPI's beta after investing in the printer business will decline to 1.28 (i.e., $.80 \times 1.4 + .20 \times .8 = 1.28$), and with this lower beta, HPI's required rate of return on its equity (and WACC because the firm is assumed to be all-equity-financed) will decline to $5\% + 1.28 \times 5\% = 11.4\%$. (See the Technical Insight box on portfolio betas.)

One way to assess whether the printer business makes sense is to ask whether HPI, with the addition of the printer business, generates an expected rate of return that exceeds the 11.4% rate of return that will be the firm's cost of capital if it enters the printer business. To calculate whether this is indeed the case, we combine the 10.5% expected

Review 1: HPI



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rate of return on the printer business (which was based on the cost of entering into the business) with the expected rate of return on the firm's existing business (which is 12% based on the firm's current market value). With the addition of the new project, the firm will generate an expected return of $.80 \times 12\% + .20 \times 10.5\% = 11.7\%$, which exceeds the required rate of return of 11.4%. This implies that the addition of the printer business will cause HPI's stock price to increase enough to drive down its future expected rate of return to its required return of 11.4%. In other words, the printer business is a value-increasing investment.

In practice, valuing an investment project by calculating the firm's WACC, both with and without the evaluated project, would be quite tedious. However, these calculations are not necessary. We can arrive at exactly the same conclusion by evaluating the project in isolation, using a discount rate that reflects the project's beta of .80. Specifically, we should evaluate the project by discounting its expected cash flows at a rate estimated using the CAPM, where the risk-free rate and market risk premium are both equal to 5%—that is, $5\% + .8 \times 5\% = 9\%$.

It should be noted that the preceding example is relatively straightforward because HPI and the printer project are assumed to be all-equity-financed. Debt financing, because of its tax advantage, is assumed to be a less expensive source of capital, so we also have to account for the extent to which the project contributes to the firm's ability to raise debt financing. As our discussion in the rest of the chapter illustrates, the consideration of debt financing definitely adds a few wrinkles to our analysis.

**IRR > required rate of
return
(11.7% > 11.4%)
Accept!**

Problem 1

5-4 CALCULATING A FIRM'S WACC AND PROJECT WACC Amgel Manufacturing Company's current capital structure is comprised of 30% debt and 70% equity (based on market values). Amgel's equity beta (based on its current level of debt financing) is 1.20, and its debt beta is 0.29. Also, the risk-free rate of interest is currently 4.5% on long-term government bonds. Amgel's investment banker advised the firm that, according to her estimates, the market risk premium is 5.25%.

- a. What is your estimate of the cost of equity capital for Amgel (based on the CAPM)?
- b. If Amgel's marginal tax rate is 35%, what is the firm's overall weighted average cost of capital (WACC)? *Assumption: Cost of debt = 6%*
- c. Amgel is considering a major expansion of its current business operations. The firm's investment banker estimates that Amgel will be able to borrow up to 40% of the needed funds and maintain its current credit rating and borrowing cost. Estimate the WACC for this project.

Problem 1

a. CAPM

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

- $k_e = 4.5\% + 1.2 \times 5.25\% = 10.8\%$

b. WACC

$$WACC = k_d(1 - T)w_d + k_p w_p + k_e w_e$$

- $WACC = 6\% \times (1 - 35\%) \times 30\% + 10.8\% \times 70\% = 8.73\%$

Problem 1

c. Unlevered beta

$$\beta_{\text{Unlevered}_j} = \frac{\beta_{\text{Equity}_j} + \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)}{1 + \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)}$$

$$\beta_{\text{Equity}_j} = \beta_{\text{Unlevered}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)$$

- $\beta_{\text{unlevered}} = \frac{1.2 + 0.29 \times \frac{30\%}{70\%}}{1 + \frac{30\%}{70\%}} = 9.27\%$; $\beta_{\text{equity}} = 9.27\% \times \left(1 + \frac{40\%}{60\%} \right) - 0.29 \times \frac{40\%}{60\%} = 1.35$
- $k_e^* = 4.5\% + 1.35 \times 5.25\% = 11.6\%$
- $WACC^* = 6\% \times (1 - 35\%) \times 40\% + 11.6\% \times 60\% = 8.52\%$

Problem 1

- **Takeaway:** This analysis presumes that the cost of debt financing is "sticky" in the sense that it varies in a **discrete fashion** with the firm's bond rating. In other words, even though the firm has increased its use of financial leverage from 30 to 40%, the cost of debt and the debt beta do not change. On the other hand, the cost of equity does change since the higher leverage implies a higher equity beta.

Review 2: Divisional WACC



Figure 5-2 Choosing the Right WACC—Discount Rates and Project Risk

Method		Description	Advantages	Disadvantages	When to Use
Firm WACC (discussed in Chapter 4)		Estimate the WACC for the firm as an entity and use it as the discount rate on <i>all</i> projects.	• Is a familiar concept to most business executives • Minimizes estimation costs, as there is only one cost of capital calculation for the firm • Does not create influence cost issues	• Does not adjust discount rates for differences in project risk • Does not provide for flexibility in adjusting for differences in project debt capacities	• Projects are similar in risk to the firm as a whole • Using multiple discount rates creates significant influence costs
Method #1: Divisional WACC		Estimate WACC for individual business units or divisions of the firm. Use these estimates as the only discount rates within each division.	• Uses division-level risk to adjust discount rates for individual projects • Entails minimal influence costs within divisions	• Does not capture intra-division risk differences in projects • Does not account for differences in project debt capacities within divisions • Potential influence costs associated with the choice of discount rates across divisions • Difficult to find single-division firms to proxy for divisions	• Individual projects within each division have similar risks and debt capacities • Discount rate discretion creates significant influence costs within divisions <u>but not between divisions</u>
Method #2: Project-specific WACC	Project financing	Estimate WACC for each individual project, using the capital costs associated with the actual financing package for the project.	• Provides a unique discount rate that reflects the risks and financing mix of the project	• Market proxies for project risk may be difficult to find • Creates the potential for high influence costs as managers seek to manipulate to get their pet projects accepted • Capital structure weights are problematic, as equity value of the project is unobservable	• The project is financed with nonrecourse debt • The costs of administering multiple discount rates are not too great
	Corporate financing	Estimate WACC for each individual project, using the capital costs associated with the debt capacity of the project.	• Provides a unique discount rate that reflects the risks and financing mix of the project	• All of the above, plus: • Project debt capacity must be allocated because it is not readily observed	• The project is of such significance that it has a material impact on the firm's debt capacity

Review 2: Divisional WACC

Table 5-1 ExxonMobil Divisions (Business Units)—2008 Operating Results

	Upstream—Oil and Gas Exploration and Development	Downstream— Refining Oil and Gas for Energy Use	Chemicals— Conversion of Crude Into Plastics and Other Nonenergy Products
Earnings	\$35.402 billion	\$8.151 billion	\$2.957 billion
Return on average capital employed*	53.6%	31.8%	20.4%
Capital expenditures	\$19.734 billion	\$3.529 billion	\$2.819 billion

Source: ExxonMobil 2008 Annual Report.

*Return on average capital employed is a performance measure ratio. From the perspective of the business segments, this return is annual business segment earnings divided by average business segment capital employed (average of beginning and end-of-year amounts). These segment earnings include ExxonMobil's share of segment earnings of equity companies, consistent with our capital employed definition, and exclude the cost of financing.

Table 5-2 Summary Divisional Costs of Capital Estimates for ExxonMobil

	Upstream—Oil and Gas Exploration and Development (SIC 1300) ^a	Downstream— Refining Oil and Gas for Energy Use (SIC 2900) ^b	Chemicals— Production of Plastics and Other Nonenergy Products (SIC 2800) ^c
Cost of equity ^d	10.17%	7.70%	8.51%
Debt ratio ^e	21.82%	4.93%	18.18%
WACC ^f	10.01%	7.78%	8.68%

Source: Ibbotson Associates, *Cost of Capital: 2009 Yearbook* (data through March 2009).

^aSIC 1300—oil and gas extraction. This group of firms is primarily engaged in (1) producing crude petroleum and natural gas, (2) extracting oil from oil sands and oil shale, (3) producing natural gasoline and cycle condensate, and (4) producing gas and hydrocarbon liquids from coal at the mine site.

^bSIC 2900—petroleum refining and related industries. This major group includes firms primarily engaged in petroleum refining, manufacturing paving and roofing materials, and compounding lubricating oils and greases from purchased materials.

^cSIC 2800—chemicals and allied products. This group includes firms that produce basic chemicals, as well as firms that manufacture products by predominantly chemical processes.

^dIbbotson reports five different estimates of the cost of equity, as we discussed in Chapter 4. However, we report the return estimate for the large composite sample, based on CAPM.

^eThe debt ratio is equal to the book value of debt divided by the sum of the book value of debt and the market value of the firm's equity. Again, this ratio reflects an average of the firms in the large-composite category.

^fWe report WACC for the large-composite group within the industry that is calculated using the CAPM-based estimate of the cost of equity.

Problem 2

5-7 DIVISIONAL WACC In 2006, Anheuser-Busch Companies Inc. (BUD) engaged in the production and distribution of beer worldwide, operating through four business segments: domestic beer, international beer, packaging, and entertainment. The domestic beer segment offers beer under Budweiser, Michelob, Busch, and Natural brands in the United States, in addition to a number of specialty beers that include nonalcohol brews, malt liquors, specialty malt beverages, and energy drinks. The international beer segment markets and sells Budweiser and other brands outside the United States and operates breweries in the United Kingdom and China. In addition, the international beer segment negotiates and administers license and contract brewing agreements with various foreign brewers. The packaging segment manufactures beverage cans and can lids for drink customers, buys and sells used aluminum beverage containers, and recycles aluminum containers. The entertainment segment owns and operates theme parks. In 2005, Anheuser-Busch reported the following segment revenues and net income:

Dollar Amount (Millions) for	Domestic Beer	International Beer	Packaging	Entertainment
2005				
Gross sales	\$10,121.00	\$864.00	\$1,831.50	\$904.40
Income before income taxes	2,293.40	70.10	120.40	215.10
Equity income	—	147.10	—	—
Net income	\$ 1,421.90	\$ 433.70	\$ 74.60	\$133.40

Assume that you have just been charged with the responsibility for evaluating the divisional cost of capital for each of the business segments.

- Outline the general approach you would take in evaluating the cost of capital for each of the business segments.
- Should the fact that \$1,156 million of the packaging segment's revenues come from internal sales to other Busch segments affect your analysis? If so, how?



Problem 2

- a. For each segment,
 - 1. Identify a set of comparable companies that have traded equity, where “comparable” is defined as operating exclusively in a similar line of business and being of similar size.
 - 2. Use standard regression techniques to compute the beta of levered equity of the comparable companies.
 - 3. Unlever the levered equity betas using standard formula (assume reasonable values for beta of debt).
 - 4. Calculate the average of the unlevered betas of comparable companies. This is a proxy for the unlevered beta of the segment.

Problem 2

5. Relever the unlevered beta of the segment using a D/E ratio that is consistent with the debt capacity of the segment.
6. Apply the CAPM to the relevered beta to get the cost of equity.
7. Apply the CAPM to the debt beta to get the pre-tax cost of debt (note: assume a constant risk-free rate, and sticky debt beta and market premium unless credit risk changes).
8. Compute the segment-WACC as the weighted average of the cost of equity and the after-tax cost of debt (note: the weights on the after-tax cost of debt is given by the ratio of book debt to (book debt + market capitalization)).

Problem 2

b. Yes, it should affect the analysis.

- Internal sales are not market-based. The 1,156 million of packaging revenue comes from selling to other Busch segments, not to outside customers. These internal transfer prices may not reflect true market value, and thus, they may overstate or understate the packaging segment's actual revenue and profitability.
- Implications for WACC analysis: The key point is that the risks of the Packaging division are **not independent**, but are largely driven by the risks of the other divisions.

Problem 2

- If the internal demand for packaging is steady and the company sets prices in a stable way, the Packaging division may look safer and less risky than a normal packaging company that sells to outside customers. But if the internal demand goes up and down a lot, or the sales are overstated, it could make the division look more risky than it really is.
- As a result, in evaluating the Packaging division's cost of capital, we may adjust for internal sales when analyzing revenue and earnings volatility.

Review 3: Project-specific WACC

1. Project financed (with nonrecourse debt financing)

- With nonrecourse debt, the project is the sole source of collateral and the debt holders have no claim to the sponsoring firm's other assets in the event of default.
- A critical feature of project financing and nonrecourse debt relates to something called bankruptcy remoteness, which means that if the borrower were to go into bankruptcy, its creditors would not have any claim on the assets of the project-financed asset.
- Investments that are project-financed are very similar to an independent firm because lenders loan money based solely on the ability of the project cash flows and assets to repay the loan.

Review 3: Catalina Inc.

- It is considering the construction and operation of a \$200 million major power-production facility in south Texas.

Figure 5-3 Project Versus Firm Financing

Panel (a) Pre-Investment Corporation

Catalina Inc.
Balance Sheet
December 31, 2014

Assets \$1,200 million

Bonds	\$ 600 million
Equity	<u>600 million</u>
Total	\$1,200 million

Panel (b) Alternative #1—Corporate Finance

Catalina Inc.
Balance Sheet
December 31, 2014

Assets	\$1,200 million
Project	<u>200 million</u>
Total	\$1,400 million

Bonds	\$ 760 million
Equity	<u>640 million</u>
Total	\$1,400 million

Panel (c) Alternative #2—Project-Financing Alternative

Catalina Inc.
Balance Sheet
December 31, 2014

Assets	\$1,200 million
Equity _{Project}	<u>40 million</u>
Total	\$1,240 million

Debt	\$ 600 million
Equity	<u>640 million</u>
Total	\$1,240 million

Power Project
Balance Sheet
December 31, 2014

Assets \$200 million

Debt	\$ 160 million
Equity	<u>40 million</u>
Total	\$ 200 million



Review 3: Catalina Inc.

Estimating Project Equity Value Directly—the Flow-Through-to-Equity Model The focus of project-financed investments is typically on the equity invested in the project, which is valued with the **flow-through-to-equity model**. This approach is simply a discounted cash flow model that focuses directly on the valuation of the equity invested in the project rather than on the value of the project as a whole. Using this approach, we calculate the present value of the equity free cash flows (equity FCFs) using the project's cost of equity capital and compare the estimated equity value with the equity invested in the project.⁷ Note that this approach does not require the calculation of the project's WACC but relies on the cost of equity and equity FCFs.

Recall from Chapter 4 that equity free cash flow, or equity FCF, can easily be computed from project FCF by recalling that a project's FCF is equal to the sum of the cash flows available to be paid to the firm's creditors (creditor cash flows) and owners (equity FCF), that is:

$$\text{Project FCF} = \text{Creditor Cash Flows} + \text{Equity FCF}$$

so that

$$\text{Equity FCF} = \text{Project FCF} - \text{Creditor Cash Flows}$$

Review 3: Catalina Inc.

So what are the cash flows paid to, or received from, the firm's creditors? There are two types of cash flows to consider: interest and principal. Also, because interest expenses are tax-deductible, we adjust interest to reflect the interest tax savings such that after-tax interest equals the firm's interest expense less the tax savings resulting from the deduction of interest from the firm's taxable income. The principal payments can be both cash outflows (to creditors) when the firm initiates new borrowing and cash inflows (to the creditors) when it repays the loan. We can summarize the calculation of creditor cash flows as follows:

$$\begin{array}{rcccl} \text{Creditor} & & \text{Interest} & - & \text{Interest} & & \text{Principal} & - & \text{New Debt Issue} \\ \text{Cash Flows} & = & \text{Expense} & & \text{Tax Savings} & + & \text{Payments} & & \text{Proceeds} \\ & & \underbrace{\hspace{10em}} & & & & \underbrace{\hspace{10em}} & & \\ & & \text{After-Tax Interest Expenses} & & & & \text{New Debt Proceeds} & & \\ & & & & & & \text{(Changes in Principal)} & & \end{array}$$

Review 3: Catalina Inc.

Table 5-3 Projected Earnings and Cash Flows for the Catalina Power-Generation Project

Panel (a) Assumptions and Forecasts

Initial investment = \$200 million in year 0

Depreciable fixed assets = \$200 million

Depreciation is straight line on \$200 million of capital equipment (no working capital), which is depreciated toward \$0.00 in 40 years.

Capital expenditures (CAPEX) = depreciation expense = \$5 million per year

Financing mix: \$160 million in debt and \$40 million in equity

Debt financing terms and conditions:

- a. Type of debt—nonrecourse project financing
- b. Maturity—infinite

c. Terms—interest only, forever

d. Interest rate—6.52%

e. Beta (debt)—30 **Typo: 0.3**

Estimated project operating earnings

(EBIT) = \$20,109,419 per year (year 1 to infinity)

Corporate tax rate = 38%

Capital market estimates and assumptions:

- a. Risk-free rate of return on long-term US government bonds = 5.02%
- b. Market risk premium = 5.00%

$$\text{After-tax interest rate} = 6.52\% \times (1 - 38\%) = 4.04\%$$

Table 5-3 continued

Panel (b) Pro Forma Income Statements and Cash Flow Projections

Pro Forma Income Statements

Year 1 to Infinity

Earnings before interest and taxes (EBIT)	\$ 20,109,419
Less: Interest expense	(10,432,000)
Earnings before taxes (EBT)	\$ 9,677,419
Less: Taxes	(3,677,419)
Net income	\$ 6,000,000

Note: Calculation of Project FCF begins with the firm's EBIT.

Calculation of Project Free Cash Flow (FCF)

Year 0

Year 1 to Infinity

Earnings before interest and taxes	\$ 20,109,419
Less: Taxes	(7,641,579)
Net operating profit after taxes (NOPAT)	\$ 12,467,840
Plus: Depreciation expense	5,000,000
Less: CAPEX	\$ (200,000,000)
Project free cash flow (FCF)	\$ (200,000,000)

Note: No interest or principal payments are considered when computing project FCF.

Calculation of Equity Free Cash Flow

Year 0

Year 1 to Infinity

Project FCF	\$ (200,000,000)	\$ 12,467,840
Less: Interest expense	—	(10,432,000)
Plus: Interest tax savings	—	3,964,160
Equals: After-tax interest expense	—	(6,467,840)
Less: Principal payments on debt	—	—
Plus: Proceeds from new debt offerings	\$ 160,000,000	—
Equals: Net proceeds from debt financing	\$ 160,000,000	—
Equity FCF	\$ (40,000,000)	\$ 6,000,000

Note: To get equity FCF, we start with project FCF, then net out cash flows paid to creditors (after-tax interest), subtract principal payments, and add new debt issue proceeds from project FCF.

Table 5-4 Estimating Project Value and the Cost of Equity for a Power-Generation Project

Panel (a) Estimating the Unlevered Project Beta Using Proxy Firms

Company Name	Levered Equity Betas	Debt/Equity Ratio	Assumed Debt Betas	Unlevered Equity Betas
Ameren Corp. (AEE)	0.7600	121.98%	0.30	0.5072
American Electric Power (AEP)	0.6000	102.38%	0.30	0.4482
Dominion Resources (D)	0.5500	76.77%	0.30	0.4414
Duke Energy Corp. (DUK)	0.4600	73.94%	0.30	0.3920
PPL Corp. (PPL)	0.6710	64.19%	0.30	0.5260
Average				0.4630

Panel (b) Iterative Procedure for Estimating the Cost of Equity, Equity Value, Project Value, and Project Finance WACC

(A) Iteration Steps	(B) Unlevered Equity Beta	(C) Debt/Equity Ratio	(D) Levered Equity Beta	(E) Estimate of the Cost of Equity	(F) Estimated Equity Value
Book weights	0.4472	4.0000	1.0361	10.200%	\$58,821,161
Iteration 1	0.4472	2.7201	0.8477	9.258%	\$64,806,671
Iteration 2	0.4472	2.4689	0.8107	9.073%	\$66,127,493
Iteration 3	0.4472	2.4196	0.8034	9.037%	\$66,393,102
Iteration 4	0.4472	2.4099	0.8020	9.030%	\$66,445,489
Iteration 5	0.4472	2.4080	0.8017	9.029%	\$66,455,782
Iteration 6	0.4472	2.4076	0.8017	9.028%	\$66,457,802

Table 5-4 continued

Panel (c) Equations Used to Estimate Project Value and the Project's Cost of Equity

$$(1) \quad \beta_{\text{Equity}_j} = \beta_{\text{Unlevered}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)$$

where β_{Debt} is the estimated debt beta for the project.⁸

$$(2) \quad \text{Levered Cost of Equity } (k_e) = \text{Risk-free Rate } (r_f) + \left(\frac{\text{Levered Equity Beta } (\beta_{\text{Equity}})}{\text{Beta } (\beta_{\text{Equity}})} \right) (\text{Market Risk Premium})$$

$$(3) \quad \text{Project Equity Value} = \frac{\text{Equity Free Cash Flow (EFCF)}_t}{\text{Levered Cost of Equity } (k_e)}$$

$$\beta_{\text{levered}} = 0.4630 \times \left(1 + \frac{160,000,000}{40,000,000} \right) - 0.3 \times \left(\frac{160,000,000}{40,000,000} \right) = 1.0361$$

$$k_e = 5.02\% + 1.0361 \times 5\% = 10.2\%$$

$$EV = \frac{6,000,000}{10.2\%} \approx 58,821,161$$

Diff=5,985,510



Diff=2,020 ≈ 0

$$\begin{aligned} \beta_{\text{levered}} &= 0.4630 \times \left(1 + \frac{160,000,000}{58,821,161} \right) - 0.3 \\ &\times \left(\frac{160,000,000}{58,821,161} \right) = 0.8477 \end{aligned}$$

- **Takeaway:** Corporate financing uses a **fixed** capital structure to calculate WACC and value the firm directly, while project financing requires **iteration** because the project's equity value affects its capital structure, which in turn influences the cost of equity used in valuation.
- In corporate financing, we already know the company's capital structure (e.g., 60% equity, 40% debt), so we can directly use WACC to value the firm. In project financing, we only know the debt amount (e.g., \$160M), but the equity value is unknown and affects the debt-to-equity ratio, which changes the cost of equity — so we must iterate until the estimated equity value leads to a consistent discount rate.

Review 3: Catalina Inc.

Calculating Project Value and Project Finance WACC Now that we have calculated the equity value, it is straightforward to value the project by simply adding the estimated value of the project's equity (from the sixth iteration in panel [b] of Table 5-4), \$66,457,802, to the project's debt, \$160 million, to arrive at a project value that equals \$226,457,802. Using this value, we can calculate project finance WACC as the weighted average of the after-tax costs of debt and equity, and we can use as weights the estimated value of the project's equity and the face value of the firm's debt:⁹

	Values	Value Weights	Capital Costs	Product
Debt	\$ 160,000,000	0.7065	4.04%	2.85%
Equity	66,457,802	0.2935	9.03%	2.65%
	\$ 226,457,802	1.0000		

$$\text{Project finance WACC} = \underline{\underline{5.51\%}}$$

Problem 3

5-8 PROJECT-SPECIFIC WACC In 2014, worldwide capacity of wind-powered generators was growing. Wind power still constitutes a small fraction of worldwide electricity use, but in Denmark, wind power provides over 20% of electricity. Moreover, the use of wind power has grown rapidly in recent years. The key feature of wind power is its heavy capital intensity and low ongoing costs (the wind is free). Combined with advances in wind power generation technology, this alternative power source appears to be one of the most promising clean energy sources for the future.

Gusty Power Company (GPC) of Amarillo, Texas, builds and operates wind farms that produce electrical power, which is then sold into the electrical grid. GPC has recently been contacted by the Plains Energy Company of Plainview, Texas, to construct one of the largest wind-power farms ever built. Plains Energy is an independent power producer that normally sells all the electrical power it produces back to the power grid at prevailing market prices. In this instance, however, Plains has arranged to sell all of its production to a consortium of electric power companies in the area for a long-term, contractually set price. Assume that the project generates perpetual cash flows (annual project FCF) of \$8,000,000, debt principal is never repaid, capital expenditures (CAPEX) equal depreciation in each period, and the tax rate is 35%.

The project calls for an investment of \$100 million, and Plains has arranged for an \$80 million project loan that has no recourse to Plains's other assets. The loan carries a 7% rate of interest, which reflects current market conditions for loans of this type. Plains will invest \$10 million in the project, and the remainder will be provided by two local power companies that are part of the consortium that will purchase the electrical power the project produces.

Assumption: cost of equity = 9.94%

Problem 3

- a. What is the project-specific WACC, calculated using the book values of debt and equity as a proportion of the \$100 million cost of the project? Given this estimate of the WACC, what is the value of the project?
- b. Reevaluate the project-specific WACC using your estimate of the value of the project from part a above as the basis for your weights. Use this WACC to reestimate the value of the project, and then recalculate the weights for the project finance WACC using this revised estimate of WACC. Repeat this process until the value of the project converges to a stable value. What is the project's value? What is the project finance WACC?
- c. Reevaluate the project-specific WACC and project value where the contract calls for a 2.5% rate of increase in the project free cash flow beginning in the second year of the project's life.

Problem 3

- a. $WACC = \frac{80,000,000}{100,000,000} \times 7\% \times (1 - 35\%) + \left(1 - \frac{80,000,000}{100,000,000}\right) \times 9.94\% = 5.63\%$
- $Enterprise\ value = \frac{8,000,000}{5.63\%} = 142,146,410.80$
 - $Value\ of\ equity = 142,146,410.80 - 80,000,000 = 62,146,410.80$
 - $DIFF = 20,000,000 - 62,146,410.80 = -42,146,410.80$
- b. $WACC_{Iteration\ 2} = \frac{80,000,000}{142,146,410.80} \times 7\% \times (1 - 35\%) + \left(1 - \frac{80,000,000}{142,146,410.80}\right) \times 9.94\% = 6.91\%$
- $Enterprise\ value_{Iteration\ 1} = \frac{8,000,000}{6.91\%} = 115,832,776.85$

Problem 3

- $Value\ of\ equity_{Iteration\ 2} = 115,832,776.85 - 80,000,000 = 35,832,776.85$
- $DIFF = 62,146,410.80 - 35,832,776.85 = 26,313,633.95$
- After 30 iterations, the imputed value of equity converges to the trial value of equity.
- $WACC_{Iteration\ 33} = \frac{80,000,000}{123,863,179.12} \times 7\% \times (1 - 35\%) + \left(1 - \frac{80,000,000}{123,863,179.12}\right) \times 9.94\% = 6.46\%$
- $Enterprise\ value_{Iteration\ 33} = \frac{8,000,000}{6.46\%} = 123,863,179.12$
- $Value\ of\ equity_{Iteration\ 33} = 123,863,179.12 - 80,000,000 = 43,863,179.12$
- $DIFF = 43,863,179.00 - 43,863,179.12 = -0.12 \approx 0$

Problem 3

$$c. \quad WACC = \frac{80,000,000}{100,000,000} \times 7\% \times (1 - 35\%) + \left(1 - \frac{80,000,000}{100,000,000}\right) \times 9.94\% = 5.63\%$$

- $Enterprise\ value = \frac{8,000,000}{5.63\% - 2.50\%} = 255.591,054.31$
- $Value\ of\ equity = 255.591,054.31 - 80,000,000 = 175,754,475.70$
- $DIFF = 20,000,000 - 175,754,475.70 = -155,754,475.70$
-

Problem 3

- Goal Seek in Excel

c. We solve this part using solver
Project FCFs are growing at 2.5%

			2.50%			
Iteration Number	Book debt	Trial value of equity	WACC	Value of project based on WACC	Imputed value of equity	diff
1	\$ 80,000,000.00	\$ 20,000,000.00	5.63%	\$ 255,754,475.70	\$ 175,754,475.70	-155754475.70

Project WACC

Project Value

Goal Seek ? X

Set cell: ↑

To value:

By changing cell: ↑

OK Cancel

c. We solve this part using solver
Project FCFs are growing at 2.5%

			2.50%			
Iteration Number	Book debt	Trial value of equity	WACC	Value of project based on WACC	Imputed value of equity	diff
1	\$ 80,000,000.00	\$ 85,483,870.97	7.33%	\$ 165,483,870.97	\$ 85,483,870.97	0.00

Project WACC

Project Value

Problem 4

5-9 PROJECT WACC USING CORPORATE FINANCING Pearson Electronics manufactures printed circuit boards used in a wide variety of applications ranging from automobiles to washing machines. In fall 2014, it considered whether to invest in two major projects. The first was a new fabricating plant in Omaha, Nebraska, which would replace a smaller operation in Charleston, South Carolina. The plant would cost \$50 million to build and incorporate the most modern fabricating and assembly equipment available. The alternative investment involved expanding the old Charleston plant so that it could match the capacity of the Omaha plant and modernizing some of the handling equipment, at a cost of \$30 million. Given the location of the Charleston plant, however, it would not be possible to modernize the plant completely due to space limitations. The end result is that the Charleston modernization alternative cannot match the out-of-pocket operating costs per unit of the fully modernized Omaha alternative.

Problem 4



Pearson's senior financial analyst, Shirley Davies, made extensive forecasts of the cash flows for both alternatives but was puzzling over what discount rate or rates she should use to evaluate them. The firm's WACC was estimated to be 9.12%, based on an estimated cost of equity capital of 12% and an after-tax cost of debt capital of 4.8%. However, this calculation reflected a debt-to-value ratio of 40% for the firm, which she felt was unrealistic for the two plant investments. In fact, conversations with the firm's investment banker indicated that Pearson might be able to borrow as much as \$12 million to finance the new plant in Omaha but no more than \$5 million to fund the modernization and expansion of the Charleston plant without jeopardizing the firm's current debt rating. Although it was not completely clear what was driving the differences in the borrowing capacities of the two plants, Shirley suspected that a major factor was that the Omaha plant was more cost effective and offered the prospect of much higher cash flows.

- a. Assuming that the investment banker is correct, use book value weights to estimate the project-specific costs of capital for the two projects. (*Hint: The only difference in the WACC calculations relates to the debt capacities for the two projects.*)
- b. How would your analysis of the project-specific WACCs be affected if Pearson's CEO decided to delever the firm by using equity to finance the better of the two alternatives (i.e., the new Omaha plant or the Charleston plant expansion)?

Problem 4

- a. WACC
$$WACC = k_d(1 - T)w_d + k_pw_p + k_ew_e$$
- $WACC_{Omaha} = 4.8\% \times \frac{12}{50} + 12\% \times \frac{38}{50} = 10.27\%$
 - $WACC_{Charleston} = 4.8\% \times \frac{5}{30} + 12\% \times \frac{25}{30} = 10.80\%$
- b. If Pearson's CEO uses 100% equity financing, both projects would have a WACC equal to the cost of equity at 12%, eliminating the benefit of cheaper debt. This would raise the cost of capital for both projects, especially Omaha, reducing their present values.

- *Thank you for your time and attention!*